Project Plan

RUNTIME TERROR

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Table of contents

1. Introduction

- 1.1 Purpose of Plan Document
- 1.2 Project Scope and Objectives
    - 1.2.1 Statement of Scope
- 1.3 Major Functions
- 1.4 Overview of Document

2. Project Organization

- 2.1 Process Model
- 2.2 Team Structure
- 2.3 Communication
- 2.4 Reporting

3. Project Resources

- 3.1 Hardware and Software

4. Risk Management

- 4.1 Likely Risks
- 4.2 Risk Management Plan
- 4.3 Risk Table

5. Schedule

- 5.1 Task List and Resource Allocation
- 5.2 Timeline
- 5.3 Task Network Diagram
- 5.4 Objective Grading Sheet

6. Tracking and Control Mechanics

- 6.1 Progress Tracking
- 6.2 Quality Control
- 6.3 Scope Control
- 6.4 Schedule Control
- 6.5 Performance Metrics

7. Glossary
8. References

1. Introduction

1. Purpose of Plan Document

-The purpose of this plan document is to break up our advising projects so it's easier for the team to understand what is going on

1.2. Project Scope and Objectives

1.2.1 Statement of Scope-

The Student Academic progress application designed to support academic advising and student success. The system will allow students to track their academic progress, and monitor degree completion. Students will be able to review student records. This application will focus strictly on academic progress tracking and setting up a schedule to make before you go to an academic advisor. The application is intended for use from a single student and will store and manage the students student academic records securely.

1.3. Major Functions -

Store and display academic history, track completed and in-progress courses, calculate and display GPA, track total earned credits, define degree requirements for a CMPS major, and display completion percentage.

1.4 Overview of document- This document outlines the complete project plan for the Student Academic Progress system. It provides a structured explanation of the system's purpose, scope, organization, schedule, risk management strategy, and evaluation criteria.

The document begins with an introduction describing the purpose of the project and defining the system's scope and major functions. It then explains the project organization, including the development process model (Agile methodology), team structure, communication methods, and reporting procedures.

Following that, the document details project resources, including personnel, hardware, and software requirements. The risk management section identifies potential technical, managerial, and user-related risks, along with strategies to reduce or prevent those risks.

The schedule section outlines task allocation, timeline planning, and the task network diagram used to organize workflow. Tracking and control mechanics describe how progress, quality, and scope will be monitored throughout development. An objective grading sheet explains how the project's success will be evaluated.

The document concludes with a glossary of key terms and a list of references used in planning and development.

2. Project Organization

This part is a summary of how we will accomplish this project using agile methodology, as well as our structure and communication.

2.1.Process Model- We are using the Agile method since it allows us to make prototypes and go back and change things without it being detrimental to our project

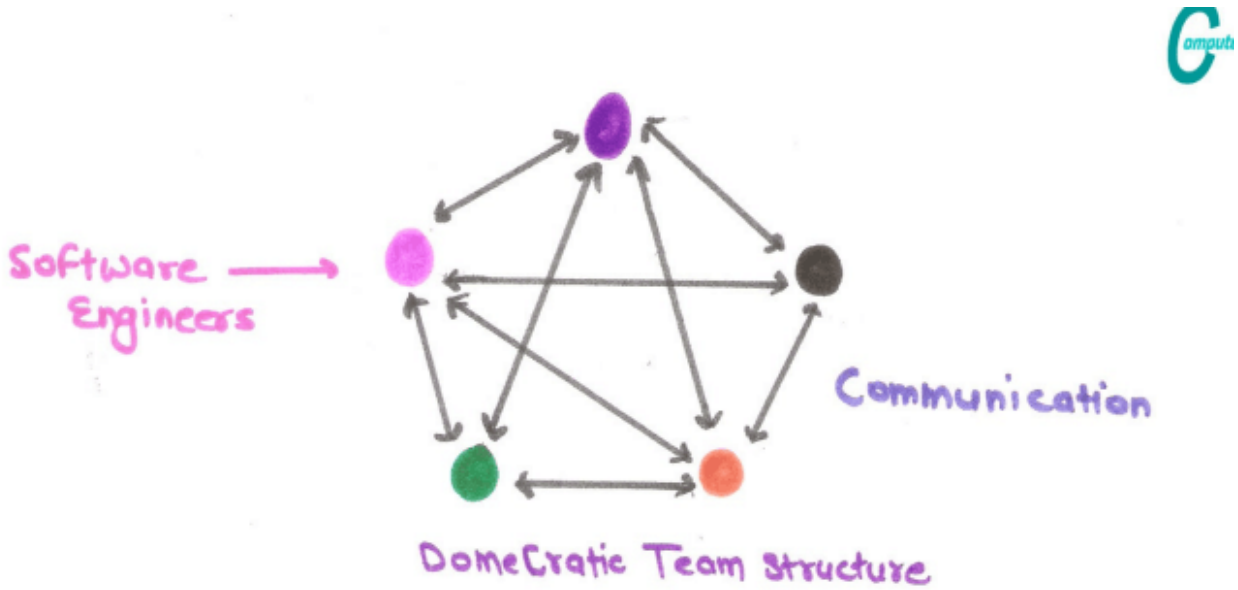
2.2.Team Structure

We are using a democratic team structure where there is no hierarchical set up, everyone has the same workload and has a say in anything we do as a team across the whole process.

- The front end is in charge of the visual aspect of our website. It will look similar to degree works with a progress tracker located at the top of the home page along with past class history and in progress classes. The front end will also display two more tabs with a “registration” tab and a “schedule” tab.
- The backend is in charge of the code that allows everything to run smoothly in order for the front end to work. The backend will be in charge of a gpa tracker as well as allowing you to switch between tabs and allow you to enter a CRN number for each class and display it in the schedule tab.
- The project leader focuses on project scope, timeline, budget, and team coordination. The leader makes sure the team has one individual idea on what the application should look like, meetings are also conducted by the leader and receives information from the customer to relay to the team.
- Data Base that makes sure we have all required information by extracting it from the 2026 list of classes on the MSU website. The database will need the CRN of each available class, subject, name of class, and the meeting time in order to have an accurate program(in our case a database of classes).
- QA is in charge of our quality assurance that debugs and makes sure our program runs effectively. This will be done by each member of the group by doing a peer review of each other's section.(Calculating GPA, Display of classes, checking each tab, and making sure it is user friendly for every person to use).

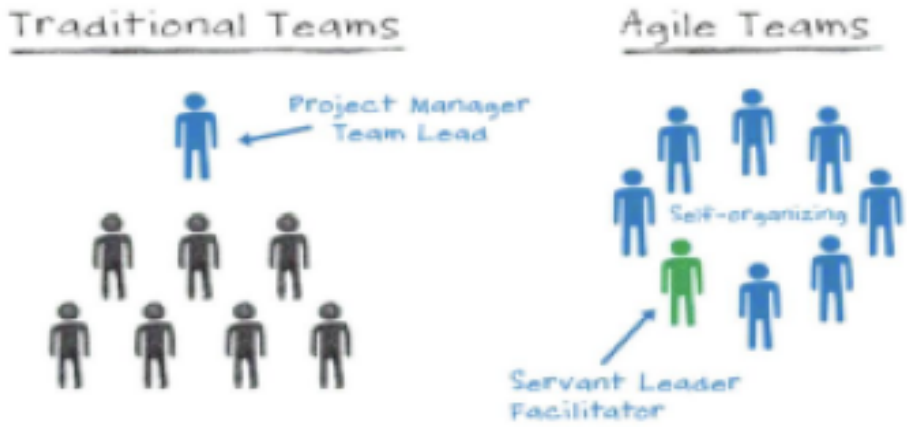
2.3.Communication- IMessage Group chats to discuss meeting times, in person meetings outside of class to talk about design of the application and to break up who is in charge of each part (at library), Email to send documents to the team leader in order to enter in Github, and Github to keep track of any work and documents for every member to see as well as keeping our work in the cloud

2.4.Reporting: Twice a week in class or out of class on Fridays after senior seminar.



The table above shows a democratic team structure

This table shows a diagram of an agile team



3. Project Resources

This states the people needed to accomplish the project and who will be using it to get this project running, as well as the hardware and software used.

People:

Project Leader (Carlos Astudillo)– Oversees scope, timeline, coordination, and ensures deliverables are met.

Frontend Developer(Jacob Lawler, Mekhi Propere) – Designs and implements the user interface and dashboard views.

Backend Developer(Kevin Kangethe, James Odjewuyi) – Develops business logic, GPA calculations, alerts, and system integration.

Database Administrator/Developer(Carlos Astudillo) – Designs and manages the academic records database (students, courses, majors, requirements).

QA Tester(Everyone) – Tests system functionality (GPA calculation accuracy, alerts, degree audit results, dashboard displays).

Students (End Users) – Use the system to track academic progress and provide usability feedback.

3.1. Hardware and Software- hardware would include laptops and a device to present the project. Software would include GitHub, Visual Studio, python, C sharp, and c++.

4. Risk Management

This states every possible risk and how we can manage these issues.

4.1.Likely Risks- incorrect gpa calculations, degree audit errors, integration difficulties, missed deadlines, miscommunication among team, incorrect data, incorrect alerts for at risk students

4.2.Risks Management Plan- conduct code reviews, data validation, provide simplicity for the student, hold meetings, and split up work evenly.

4.3 Risk table 1

5. Schedule

This schedule will be on Asana.

5.1.Task List and Resource allocation

- Requirements gathering to define features and scope.
- System design to create wireframes
- Database development to build tables for students, classes, and majors. • Backend development for gpa, degree audit, and alerts.
- Frontend development, dashboard, and student view of classes. • QA- Test GPA, alerts, and credit tracking

- Debugging and revisions to fix issues.
- Final review

5.2. Time Line
5.3. Task Network Diagram
5.4. Objective Grading Sheet

4.3. Table 1 describes a handful of risks along its likelihood and mitigation

Risks	Likelihood	Mitigation
Incorrect GPA Calculations	Medium	Implement unit tests for GPA formulas, conduct peer code reviews, and verify results using sample student records.
Degree Audit Errors	Medium	Validate degree requirements with official curriculum data and perform multiple test cases for different course combinations.
integration Difficulties Between Frontend, Backend, and Database	Medium	Use regular integration testing, maintain clear API structures, and schedule integration checkpoints during development.
Missed Project Deadlines	Medium	Follow Agile sprint planning, monitor tasks weekly, and redistribute work if deadlines begin slipping.
Team Miscommunication	Medium	Hold regular meetings, maintain group chat communication, and document updates on GitHub so everyone has access.
Incorrect Data from Course List	Low	Validate course data pulled from the MSU class list and cross-check CRN numbers, course titles, and times.
Database Errors or Data Loss	Low	Use backups, version control for database structure, and validate data inputs before storage.
User Interface Confusion	Medium	Conduct user testing with students to ensure dashboards and navigation are clear and intuitive.
System Bugs During Final Demo	Medium	Perform extensive QA testing, maintain a bug tracking log, and allocate time for debugging before final presentation.

6. Tracking and Control Mechanics

Tracking and Control Mechanics describe how the team will monitor progress, maintain quality, and ensure the project stays within scope, time, and performance expectations.

6.1. Progress Tracking

- Weekly Status Meetings – The team will meet once per week to review completed tasks, upcoming tasks, and any blockers.
- Sprint Reviews (Agile Method) – At the end of each sprint, the team will review completed features (GPA calculator, degree audit, alerts, etc.).
- Task Board (Kanban or To-Do List) – Tasks will be organized into categories such as:
    - To Do
    - In Progress
    - Testing
    - Completed
- Milestone Checkpoints – Major deliverables (Database Setup, Backend Complete, Frontend Complete, Testing Complete) will be reviewed against the timeline.

6.2. Quality Control

- Code Reviews – Team members will review each other's code to catch errors early.
- QA Testing – The QA member will test:
    - GPA calculation accuracy
    - Credit totals
    - Degree audit results
    - Alert system functionality
- User Testing – Test students/advisors will check usability and clarity of dashboards.
- Bug Tracking Log – Identified issues will be documented and assigned for correction.

6.3. Scope Control

- All new feature requests must be reviewed by the Project Leader. • Changes must align with the original purpose: academic progress tracking and advising.
- Any scope changes must be documented before implementation to prevent scope creep.

6.4. Schedule Control

- Compare weekly progress to the original timeline. •

If delays occur:

- Reassign tasks if necessary
- Adjust sprint priorities
- Increase collaboration sessions

6.5. Performance Metrics

The project will be evaluated using measurable outcomes:

• Accurate GPA calculation (100% formula correctness) •
Correct credit tracking
• Accurate degree completion percentage
• Proper alert generation for at-risk students
• Minimal system errors at final testing

7. Glossary

This glossary is every major word used for this project.

Agile Methodology – A project management approach that uses short development cycles (sprints) to allow flexibility, continuous improvement, and iterative development.

Alert System – A feature that notifies students or advisors when academic performance falls below defined thresholds (e.g., low GPA or insufficient credits).

Backend – The server-side portion of the application responsible for processing data, calculations (such as GPA), and system logic.

Credit Hours – Units that represent the amount of academic coursework completed toward a degree.

Dashboard – The main user interface screen displaying academic progress, GPA, credit totals, and alerts.

Database – A structured collection of stored data (students, courses, grades, degree requirements) used by the system.

Degree Audit – An automated process that compares completed coursework against degree requirements to determine progress toward graduation.

Frontend – The visual and interactive portion of the web application that users see and interact with.

GPA (Grade Point Average) – A numerical representation of a student's academic performance calculated from course grades and credit hours.

Milestone – A significant checkpoint in the project timeline marking the completion of

a major task.

Scope – The defined boundaries of what the project will and will not include.

Sprint – A short, time-boxed development cycle used in Agile methodology to complete specific tasks.

System Integration – The process of combining frontend, backend, and database components, so they function together as one system.

User Testing – The process of having end users test the system to ensure usability and functionality meet requirements

8. References

This part is a list of all the references that were needed to complete this project outline

Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® Guide)* (7th ed.). Project Management Institute.

Beck, Kent, et al. (2001). *Manifesto for Agile Software Development*. Agile Alliance. https://agilemanifesto.org

Agile Process slides